Molecular Biology of the Cell

Chapter 3: Proteins

Cline

September 24, 2025

Outline

Outline

The Shape and Structure of Proteins

Protein Function

Protein Building Blocks

- Proteins are polymers of amino acids linked by **peptide bonds**.
- The **polypeptide backbone** is the repeating sequence of atoms.
- The 20 common amino acids each have a unique **side chain** that determines their properties (acidic, basic, polar, nonpolar).

Figure 3-2: The 20 amino acids.

Protein Building Blocks

Forces Driving Folding

- The 3D conformation is stabilized by weak **noncovalent bonds** (hydrogen bonds, electrostatic attractions, van der Waals).
- The **hydrophobic clustering force** is a major driver, pushing nonpolar side chains into the protein's core.
- The final shape minimizes free energy (G).

Figure 3-5: Hydrophobic forces in folding.

Forces Driving Folding

Folding Dynamics

- **Denaturation** and **renaturation** show that the primary sequence dictates the final structure.
- **Molecular chaperones** assist folding and prevent aggregation.
- A folded protein exists as an ensemble of related **conformers**.

Figure 3-12: Conformers of ubiquitin.

Folding Dynamics

Secondary Structures

- **α-helix**: A rigid, right-handed cylinder stabilized by H-bonds between every fourth amino acid.
- **β-sheet**: A rigid, pleated structure formed by H-bonds between adjacent strands (can be parallel or antiparallel).
- **Coiled-coil**: Two or more α-helices wrapped around each other.

Figure 3-6: α helix and β sheet.

Secondary Structures

Structural Hierarchy & Visualization

- **Primary**: Amino acid sequence.
- **Secondary**: α-helices and β-sheets.
- **Tertiary**: Full 3D shape of one chain.
- **Quaternary**: Arrangement of multiple subunits.
- Visualization methods include backbone, ribbon, wire, and space-filling models.

Figure 3-9: Protein structure representations.

Structural Hierarchy & Visualization

Modular Units and Evolution

- A **protein domain** is an independently folding segment.
- **Protein families** share similar sequences and structures (e.g., serine proteases).
- Evolution creates new proteins via **domain shuffling**.

Figure 3-15: Domain shuffling.

Modular Units and Evolution

From Subunits to Large Structures

- Proteins assemble into dimers, tetramers (e.g., hemoglobin), filaments (e.g., actin), and fibers (e.g., collagen).
- **Disulfide bonds** covalently reinforce extracellular protein structures.
- **Viral capsids** are complex assemblies protecting the viral genome.

Figure 3-27: Viral capsid structure.

From Subunits to Large Structures

Pathological Aggregates

- **Amyloid fibrils** are insoluble β-sheet aggregates linked to diseases like Alzheimer’s.
- **Prion diseases** are caused by an infectious, misfolded protein (PrP*) that propagates its misfolded state.

Figure 3-33: Prion disease mechanism.

Pathological Aggregates

Outline

The Shape and Structure of Proteins

Protein Function

Binding Specificity

- Function depends on binding to **ligands** at a specific **binding site**.
- Specificity arises from molecular complementarity and many weak noncovalent bonds.
- **Antibodies** are a prime example of highly specific binding.

Figure 3-40: Antibody structure.

Binding Specificity

Accelerating Reactions

- Enzymes lower the **activation energy** of a reaction.
- The **active site** binds the substrate and facilitates its conversion to a transition state.
- **Enzyme kinetics** are described by **Vmax** and **Km**.

Figure 3-45: Enzymes lower activation energy.

Accelerating Reactions

Allosteric Regulation

- Activity is controlled by molecules binding to a **regulatory site**.
- **Feedback inhibition** is a common negative feedback loop.
- **Cooperative allosteric transitions** in multisubunit proteins allow for switch-like responses.

Figure 3-57: Cooperative allostery.

Allosteric Regulation

Covalent Modifications

- **Protein phosphorylation** by kinases and phosphatases acts as a molecular switch.
- **GTP-binding proteins** (like Ras) cycle between active (GTP-bound) and inactive (GDP-bound) states.
- **Ubiquitylation** marks proteins for degradation or other regulatory functions.

Figure 3-58 & 3-63: Phosphorylation and GTP-binding.

Covalent Modifications

Cellular Compartments Without Membranes

- Large, dynamic assemblies of proteins and RNA that form through **liquid-liquid phase separation**.
- Held together by weak, multivalent interactions, often involving **intrinsically disordered proteins (IDPs)**.
- Concentrate molecules to create specialized biochemical environments (e.g., nucleolus).

Figure 3-78: Fusing nucleoli.

Cellular Compartments Without Membranes